

UrbanPulse DLQ Report

DLQ Report

Problem Statement

Capture validation failures from the injection pipelines and publish a summarized DLQ report with CSV and chart artifacts.

Design

Producer validation failures are routed to `urbanpulse.dlq`; the consumer logger aggregates them into a markdown report, CSV, bar chart, and pie chart.

Implementation

See `consumers/dlq\_logger.py`.

Code Snippets

The logger writes `dlq\_report.md`, `dlq\_report.csv`, `dlq\_report\_bar.png`, and `dlq\_report\_pie.png`.

Screenshots

See `screenshots/overview.svg`.

Results

A generated DLQ report is now available in the repository root.

Challenges

Keeping the report generation lightweight and dependency-friendly.

Future Work

Expose the DLQ metrics through Grafana or a web endpoint.

Conclusion

The DLQ workflow is now documented and produced.